

MAAT v1.8: State-Conditioned Structural Selection

Evaluation Locality, Arbitrated Fallback, and the Lessons of Gate Challenge Cycle I

MAAT Structural Selection Series – Paper 74

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Abstract

Gate Challenge Cycle I (Papers 68–73, archived at <https://doi.org/10.5281/zenodo.21062386>) produced two external negative results for MAAT in SAT: the frozen v1.7 gate did not improve CDCL decisions (Paper 69), and the preregistered local occurrence/degree residual channel was operationally degenerate under the frozen implementation (Paper 73). The diagnostic papers in between located the cause precisely. Paper 70 showed that the classical MOMS advantage is decision-local structural information; Paper 71 showed that most such information is already reconstructible from the existing supports. This note argues that the common failure mode across the cycle was not the structural signal but a still-untested *evaluation point*: every external MAAT non-support result came from a construct evaluated once per instance, competing against heuristics that re-evaluate at every decision on the current residual state. Conversely, the domains in which MAAT performs well — open quantum systems and fluid diagnostics — already evaluate the supports on the evolving state, and their strongest competitors (short-time generator baselines, spectral-tail monitors) are themselves state-local quantities. MAAT v1.8 therefore proposes *state-conditioned structural selection*: (i) the supports become functionals of the current decision state, $\Gamma_a[X_t]$, with the v1.2.1 robustness closure and the frozen v1.7 response gate unchanged except for their argument; (ii) gate activation is embedded in an *arbitrated fallback* to a strong classical default, with a predeclared no-harm bound, so that selective non-action becomes a first-class measurable outcome; and (iii) gate thresholds are calibrated by a predeclared *activation budget* rather than by signal quantiles. Two new protocol rules generalize the Cycle-I lessons to all future preregistrations: a distinctness precondition on channel constructions (from Paper 73) and a classical-reconstruction check on decision-local supports (from Paper 71). This note is a framework hypothesis. It reports no new execution, does not modify the Paper-68 gate record, does not reinterpret the Paper-69 or Paper-73 outcomes, and defers all quantitative success criteria to a future Cycle-II preregistration.

1 Status and Guardrails

This is a framework note, written after the completion and archival of Gate Challenge Cycle I and before any v1.8 execution. Its role is the same as that of the v1.7 note relative to Paper 68: it states a hypothesis and its motivation, so that a subsequent preregistration can freeze the corresponding test.

This note does not modify the frozen Paper-68 gate architecture, does not reinterpret the negative Paper-69 and Paper-73 outcomes, does not retune any frozen parameter, and does not report or anticipate any v1.8 execution result. The Paper-69 and Paper-73 records stand, permanently, as the external results for statically evaluated MAAT constructs in SAT.

If state-conditioned evaluation is useful, that must be shown in a new predeclared experiment (Gate Challenge Cycle II). It cannot be used retroactively to soften the Cycle-I record.

2 Diagnosis: The Evaluation-Locality Pattern

Placing the Cycle-I results next to the earlier internal benchmarks reveals a single pattern.

Evidence	Observation
Paper 69	The frozen v1.7 gate was evaluated once per instance (root level) and lost clearly to MOMS, which re-evaluates on the current residual formula at every decision.
Paper 70	The MOMS advantage was diagnosed as decision-local structural information, especially short-clause and occurrence pressure on the <i>current</i> state.
Paper 71	Most global local-channel candidates are affinely reconstructible from the existing support basis; adding new static channels is a closed direction.
Paper 73	The last surviving residual channel was collinear by construction ($ O_{\text{raw}} - D_{\text{raw}} \sim 10^{-10}$) and therefore inert; the tested static-channel construction became operationally degenerate.
Paper 64	Under leave-channel-family-out transfer, the strongest predictor was the <i>short-time generator baseline</i> — derivatives evaluated on the current state — not the full static defect-field model.
Paper 67	The strongest competitor to the structural warning was the high- k spectral fraction, a state-local quantity of the current snapshot.
Quantum, Fluid	The two domains where MAAT constructs perform well already evaluate H, B, S, V on the evolving state ρ_t or $\omega(\cdot, t)$. They were, in this sense, v1.8-conformant before v1.8 existed.

Table 1: Evidence rows for the evaluation-locality diagnosis. Every external MAAT loss in Cycle I involved a statically evaluated construct; every strong competitor and every internal MAAT success involved state-local evaluation.

The compressed statement of the diagnosis is:

The structural signal was not shown to be absent. The evaluation point remains the next untested degree of freedom.

Papers 71 and 73 close the “new channel” direction: the information content of the tested static instance-level MAAT descriptions appears saturated. What has never been tested externally is the same support language evaluated *where the classical heuristics evaluate*: on the current decision state.

3 v1.8 Definition

3.1 State-Conditioned Supports

Let X_t denote the current decision state of a process at decision time t . The v1.8 supports are the existing operational supports evaluated on X_t :

$$\Gamma_a[X_t], \quad a \in \{H, B, S, V\}. \quad (1)$$

The domain instantiations are:

Domain	Decision state X_t	Change relative to Cycle I
SAT/CDCL	residual formula $F _{\pi_t}$ under partial assignment π_t	new: supports move from F to $F _{\pi_t}$
Open quantum	current density matrix ρ_t	none: already state-evaluated
Fluid	current field snapshot $\omega(\cdot, t)$	none: already state-evaluated

Table 2: Domain instantiation of the decision state. That the quantum and fluid columns require no change is the strongest internal consistency argument for v1.8: the domains where MAAT already works are exactly the domains that already satisfy its definition.

The v1.2.1 robustness closure is unchanged except for its argument:

$$R_{\text{resp}}[X_t] = (H[X_t] B[X_t] V[X_t])^{1/3}, \quad (2)$$

$$R_{\text{rob}}[X_t] = \min\left(R_{\text{resp}}[X_t], (H[X_t] B[X_t] S[X_t] V[X_t])^{1/4}\right). \quad (3)$$

The frozen Paper-68 response gate is likewise reused without modification of its architecture or constants ($a_t = 0.50$, $a_x = 0.25$, $\tau = 1$, hard threshold 0.5), fed with $R_{\text{rob}}[X_t]$:

$$g[X_t] = z_R[X_t] + a_t z_t[X_t] + a_x z_x[X_t], \quad G_{\text{gate}}[X_t] = \frac{1}{1 + \exp(-g[X_t]/\tau)}. \quad (4)$$

One definitional consequence must be stated explicitly, because it is a real change and not a detail: the calibration statistics (median and MAD entering z_R , and the standardizations entering z_t, z_x) are computed over the *decision states of the calibration instances*, not over the calibration instances themselves. This is forced by the definition — a per-decision gate must be normalized on the population it will act on — and it must be frozen as such in the Cycle-II preregistration.

3.2 Arbitrated Fallback with a No-Harm Bound

The honest lesson of Cycle I is that MAAT must earn activation against strong classical defaults. The v1.8 policy is therefore an arbitration, not a replacement:

$$\pi_{\text{v1.8}}(t) = \begin{cases} \pi_{\text{MAAT}}(X_t) & \text{if } G_{\text{gate}}[X_t] \geq \theta_q, \\ \pi_{\text{default}}(t) & \text{otherwise,} \end{cases} \quad (5)$$

where π_{default} is a predeclared strong domain-native policy (candidates: MOMS or VSIDS in SAT; the high- k monitor in fluids; the stationarity baseline in quantum ranking). Alongside any positive criterion, the Cycle-II preregistration must include a *no-harm bound*:

$$\text{CI}_{95}^{\text{high}}\left[\overline{\text{regret}}(\pi_{\text{v1.8}}, \pi_{\text{default}})\right] \leq \delta, \quad (6)$$

with δ predeclared. This makes *selective non-action* a first-class measurable outcome. In SAT in particular, the realistic v1.8 ambition is not “beat MOMS everywhere”; it is “know when not to act, and act only where structural information helps, at bounded cost.” A policy that satisfies Eq. (6) while activating on a predeclared fraction of decisions and improving that subset is a genuine routing result, not a consolation prize — it is the algorithm-selection direction that the SAT thread of this programme has pointed toward since the family-sensitivity findings.

3.3 Budget Calibration

Gate thresholds in the v1.7 pilot were signal quantiles (q_{75}), which the v1.7 note itself listed as an open weakness. v1.8 replaces this with a predeclared *activation budget*: choose $q \in (0, 1)$ in advance, and set

$$\theta_q = \text{the } (1 - q)\text{-quantile of } G_{\text{gate}}[X_t] \text{ over the calibration-fold decision states.} \quad (7)$$

The gate then opens on approximately a fraction q of decisions by construction. This is the matched-rate logic of the Paper-67 false-alarm calibration, transferred from warning systems to activation systems: policies are compared at equal intervention rate, so no policy wins merely by acting more often. The value of q is deliberately *not* fixed in this note; it is a Cycle-II preregistration parameter (a single value, e.g. of order $q \approx 0.25$, fixed before execution and never scanned).

4 Computational Feasibility Requirement

State-conditioned supports are only meaningful if they are affordable at decision frequency. For SAT, the occurrence, degree, and clause-length statistics entering the supports admit incremental maintenance: counters updated on clause satisfaction and clause shrinkage, with rollback on backtrack, at cost $O(\text{touched literals})$ per decision — the same cost class as VSIDS activity bumping. v1.8 makes this a *deliverable*, not a promise: any Cycle-II execution must report the measured per-decision overhead of the state-conditioned supports relative to the default policy, and the overhead must be included in the compute-cost metric rather than excluded from it. A gate that helps only when its own cost is ignored has not helped.

5 Two New Protocol Rules

Cycle I produced two methodological lessons that should bind all future preregistrations, independently of v1.8.

5.1 Distinctness Precondition (from Paper 73)

Before any execution, the pairwise correlations of all constructed channels must be computed and reported on the calibration fold. If two channels satisfy $1 - |\rho| < \varepsilon$ for a predeclared small ε , the construct is declared degenerate and the planned execution is classified as

`not_testable_as_constructed,`

which is evidence in *no* direction. Paper 73 discovered this failure honestly but only at execution time; the rule moves the check before the run, where it costs nothing and protects the interpretation.

5.2 Classical-Reconstruction Check (from Paper 71)

Paper 71 asked whether new channels are reconstructible from MAAT supports. The state-conditioned setting requires the mirrored question: are the decision-local MAAT supports reconstructible from the classical heuristics? On the calibration fold, fit the decision-local supports affinely from {MOMS, JW, VSIDS-activity} scores. If the reconstruction R^2 exceeds a predeclared ρ_* , the preregistered conclusion is

`convergent_rediscovery —`

the state-conditioned supports would then be a structural *re-derivation* of known heuristics, which is an interesting interpretive result but not a new signal, and must be reported as such. This is the sharpest falsification channel v1.8 has: it allows the hypothesis to fail in the most informative possible way.

6 Why v1.8, Not v2.0

- No v1.8 execution exists. This note precedes its own evidence, by design.
- The SAT expectation is deliberately modest: plausibly better than the inert static gate; uncertain against MOMS on SATLIB families, for which MOMS is nearly tailor-made.
- The strongest expected domains remain quantum and fluid — which require no definitional change and therefore cannot, by themselves, confirm the new part of the hypothesis.
- The arbitration component has value even under partial SAT failure, but “value under failure” is a design property, not evidence.
- The activation budget q , the no-harm bound δ , the distinctness ε , and the reconstruction threshold $\rho_\star$ are not yet frozen; freezing them is the job of the Cycle-II preregistration.

v1.8 is therefore the correct level of claim: a framework hypothesis with a precise, testable core and a predeclared path to its own refutation.

7 Version Genealogy

Version	Healed weakness	Resulting principle
v1.2.1	fifth-field double counting	R demoted to emergent closure
v1.4	scalar compression failed in SAT	multi-scale defect modes
v1.5	population balance failed in open systems	generator stationarity
v1.6	static ranking vs. search	structural search geometry
v1.7	three convergent R_{rob} -as-score failures	score $\neq$ gate
v1.8	static evaluation failed in Cycle-I SAT tests (69, 73); state-local baselines won (64, 67)	signal $\neq$ evaluation point

Table 3: Each version is a healed weakness, not an added claim. v1.8 continues the pattern: it is derived from the Cycle-I diagnostic record, not from a new idea seeking evidence.

8 Practical Rule

A v1.8 workflow, for any domain:

1. define the decision state X_t and the supports as functionals of it;
2. verify the distinctness precondition on all constructed channels;
3. run the classical-reconstruction check and predeclare its threshold;
4. freeze the activation budget q , the default policy, and the no-harm bound δ ;
5. measure and include the per-decision overhead in the cost metric;
6. compare default-only, static-gate, and state-conditioned arbitration at matched activation rate;
7. report failures, `not_testable_as_constructed`, and `convergent_rediscovery` outcomes with the same prominence as successes.

The governing methodological sentence, extending the v1.7 rule:

A structural signal is only as informative as the state it is evaluated on. v1.7 asked whether a mode should act; v1.8 asks at which state the question is even posed.

9 What v1.8 Does Not Claim

This note makes no claim that state-conditioned supports will beat MOMS, no claim that the Cycle-I record was incomplete or unfairly evaluated, no claim that quantum or fluid successes already validate v1.8 (they predate it and required no change), and no claim of universality. The correct interpretation is narrower:

v1.8 identifies the single untested degree of freedom left open by the Cycle-I diagnostics — the evaluation point — and proposes the minimal framework change that tests it, together with the guardrails that make an honest test possible.

10 Next Step

The path follows the established sequence:

Paper 74 (this note) → Paper 75 (Cycle-II preregistration)
→ Zenodo DOI → Paper 76 (execution).

Paper 75 must freeze, at minimum: the activation budget q , the no-harm bound δ , the distinctness threshold ε , the reconstruction threshold $\rho_\star$, the default policy per domain, the exact code-level definitions of the state-conditioned supports on $F|_{\pi_t}$, the decision-state calibration population, the overhead-accounting rule, datasets, budgets, baselines, and the full success/failure ladder. It must be deposited with a public timestamp before any state-conditioned solver run, exactly as Paper 68 was deposited before Paper 69.

11 Reproducibility

Repository:

<https://github.com/Chris4081/structural-selection-principle>

This note introduces no dataset and no execution. The planned protocol folder is

`experiments/maat_v18_state_conditioned_prereg/`

and will contain the machine-readable Cycle-II preregistration once frozen.

12 Conclusion

Gate Challenge Cycle I did not support two specific MAAT constructs in SAT and, in doing so, located the shared assumption they rested on: that a structural description of the instance suffices, evaluated once, while the competing heuristics evaluate the instance’s *current state* at every decision. MAAT v1.8 removes that assumption. The supports keep their definitions and their closure; the gate keeps its frozen architecture; what changes is the argument — the state on which the question is posed — together with an arbitration that makes non-action safe and measurable, and a budget calibration that makes activation rates comparable. Whether this is enough is not a matter for this note. It is a matter for Cycle II.

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